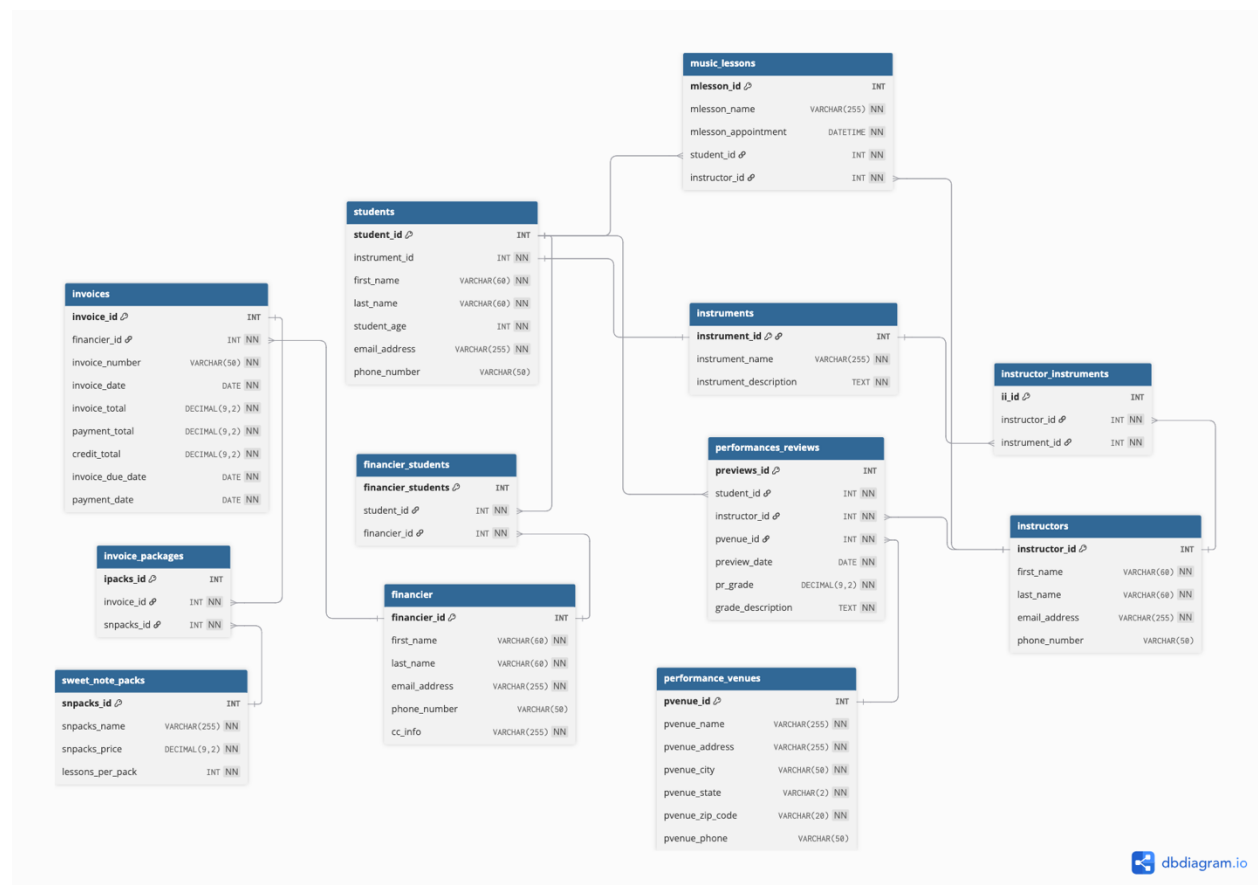


Technical Documentation

Yasmin Shawamreh | DBA210 Final Project

[Sweet Note Music Academy ERD]:

<https://dbdiagram.io/d/Sweet-Note-Music-Academy-687e7aa5f413ba3508ef4702>



Testing & Sample Queries:

Ensure all students are age 5 or above:

```
ALTER TABLE students
ADD CONSTRAINT chk_student_age CHECK (student_age >= 5);
Query OK, 11 rows affected (0.28 sec)
Records: 11 Duplicates: 0 Warnings: 0
```

Test:

```
INSERT INTO students VALUES
(DEFAULT, 1, 'Avery', 'Gitburg', 3, 'gitburg@gmail.com', '828-456-7890');
```

ERROR 3819 (HY000): Check constraint 'chk_student_age' is violated.

Find students names, their instruments and age:

```
SELECT first_name, last_name, i.instrument_name, student_age
FROM students s
JOIN instruments i ON s.instrument_id = i.instrument_id;
```

first_name	last_name	instrument_name	student_age
Emma	Goodman	Voice	17
Liam	Sommers	Drums	10
Olivia	Brown	Flute	21
Noah	Abdullah	Guitar	45
Eva	Martinez	Voice	12
Elijah	Garcia	Cello	67
Amina	Hassan	Bass Guitar	30
James	Anderson	Clarinet	18
Isabella	Thomas	Violin	12
Zahra	Abdi	Saxophone	11
Ali	Abdi	Guitar	13

11 rows in set (0.00 sec)

Retrieve all music lessons for a specific student:

```
SELECT first_name, last_name, mlesson_name,  
DATE_FORMAT(ml.mlesson_appointment, '%W, %M %D, %Y - %l:%i %p') AS  
music_lesson_appt  
FROM students s JOIN music_lessons ml  
ON s.student_id = ml.student_id  
WHERE s.first_name = 'Ali' AND s.last_name = "Abdi";
```

first_name	last_name	mlesson_name	music_lesson_appt
Ali	Abdi	Guitar	Wednesday, August 13th, 2025 - 1:30 AM
Ali	Abdi	Guitar	Sunday, August 17th, 2025 - 12:30 PM
Ali	Abdi	Guitar	Friday, August 22nd, 2025 - 12:30 PM

3 rows in set (0.00 sec)

Display upcoming music lessons for August:

```
SELECT ml.mlesson_name, DATE_FORMAT(ml.mlesson_appointment, '%W, %M  
%D, %Y - %l:%i %p') AS august_lessons,  
CONCAT(ins.first_name, ' ', ins.last_name) AS instructor_name,  
CONCAT(stu.first_name, ' ', stu.last_name) AS student_name  
FROM music_lessons ml  
JOIN instructors ins ON ml.instructor_id = ins.instructor_id  
JOIN students stu ON ml.student_id = stu.student_id  
WHERE ml.mlesson_appointment BETWEEN '2025-08-01' AND '2025-08-31'  
ORDER BY ml.mlesson_appointment ASC;
```

mlesson_name	august_lessons	instructor_name	student_name
Violin	Monday, August 4th, 2025 - 2:30 AM	Fatima Al-Farsi	Isabella Thomas
Clarinet	Monday, August 4th, 2025 - 12:30 PM	Hyeon Parker	James Anderson
Drums	Tuesday, August 5th, 2025 - 1:30 AM	Raj Patel	Liam Sommers
Voice	Tuesday, August 5th, 2025 - 12:30 PM	Leah Dubois	Emma Goodman
Bass Guitar	Wednesday, August 6th, 2025 - 1:30 AM	Mei Chen	Amina Hassan
Cello	Wednesday, August 6th, 2025 - 2:30 AM	Fatima Al-Farsi	Elijah Garcia
Cello	Sunday, August 10th, 2025 - 1:30 AM	Fatima Al-Farsi	Elijah Garcia
Guitar	Sunday, August 10th, 2025 - 3:30 AM	Jamal Washington	Noah Abdullah
Bass Guitar	Monday, August 11th, 2025 - 12:30 PM	Mei Chen	Amina Hassan
Flute	Tuesday, August 12th, 2025 - 1:30 AM	Hyeon Parker	Olivia Brown
Guitar	Wednesday, August 13th, 2025 - 1:30 AM	Jamal Washington	Ali Abdi
Saxophone	Wednesday, August 13th, 2025 - 12:30 PM	Raj Patel	Zahra Abdi
Guitar	Sunday, August 17th, 2025 - 12:30 PM	Jamal Washington	Ali Abdi
Voice	Wednesday, August 20th, 2025 - 12:30 PM	Leah Dubois	Eva Martinez
Guitar	Friday, August 22nd, 2025 - 12:30 PM	Jamal Washington	Ali Abdi
Voice	Sunday, August 24th, 2025 - 12:30 PM	Leah Dubois	Eva Martinez

16 rows in set (0.00 sec)

Find invoice for specific financier:

```
SELECT CONCAT(f.first_name, ' ', f.last_name) AS financier_name,  
i.invoice_number, i.invoice_total  
FROM invoices i  
JOIN financier f ON i.financier_id = f.financier_id  
WHERE f.first_name = 'Omar' AND f.last_name = 'Abdi';
```

```
+-----+-----+-----+  
| financier_name | invoice_number | invoice_total |  
+-----+-----+-----+  
| Omar Abdi     | 812000-401     |          900.00 |  
+-----+-----+-----+  
1 row in set (0.00 sec)
```

Find multi instrumentalist instructors & what they teach:

```
SELECT CONCAT(i.first_name, ' ', i.last_name) AS instructor_name,  
GROUP_CONCAT(DISTINCT ins.instrument_name ORDER BY ins.instrument_name  
SEPARATOR ', ') AS instruments_taught  
FROM instructors i  
JOIN instructor_instruments ii ON i.instructor_id = ii.instructor_id  
JOIN instruments ins ON ii.instrument_id = ins.instrument_id  
GROUP BY instructor_name  
HAVING COUNT(DISTINCT ii.instrument_id) > 1;
```

```
+-----+-----+  
| instructor_name | instruments_taught |  
+-----+-----+  
| Fatima Al-Farsi | Cello, Violin     |  
| Hyeon Parker    | Clarinet, Flute   |  
| Jamal Washington | Drums, Guitar, Piano, Violin |  
| Leah Dubois     | Piano, Voice      |  
| Raj Patel       | Drums, Saxophone  |  
+-----+-----+  
5 rows in set (0.00 sec)
```

Implement Views:

VIEW to display quarterly report cards:

```
CREATE VIEW view_report_card AS
SELECT
CONCAT(i.first_name, ' ', i.last_name) AS instructor_name,
CONCAT(s.first_name, ' ', s.last_name) AS student_name, ins.instrument_name AS
instrument, DATE_FORMAT(pr.preview_date, '%W, %M %D, %Y') AS performance_date,
pr.pr_grade AS grade, pr.grade_description, pv.pvenue_name AS venue_name
FROM performance_reviews pr
JOIN performance_venues pv ON pr.pvenue_id = pv.pvenue_id
JOIN instructors i ON pr.instructor_id = i.instructor_id
JOIN students s ON pr.student_id = s.student_id
JOIN instruments ins ON s.instrument_id = ins.instrument_id
ORDER BY pr.preview_date DESC;
```

Query OK, 0 rows affected (0.03 sec)

Test:

```
SELECT * FROM view_report_card;
```

instructor_name	student_name	instrument	performance_date	grade	grade_description	venue_name
Leah Dubois	Emma Goodman	Voice	Tuesday, July 15th, 2025	85.00	Needs more rehearsal. Pitch inconsistencies observed.	The One Stop at Asheville Music Hall
Leah Dubois	Eva Martinez	Voice	Thursday, July 10th, 2025	95.00	Great performance! Slight timing issue on transitions.	Asheville Music Hall
Leah Dubois	Noah Abdullah	Guitar	Friday, June 20th, 2025	95.00	Nervous start, but improved steadily.	Sly Gorg Lounge
Fatima Al-Farsi	Elijah Garcia	Cello	Tuesday, June 10th, 2025	100.00	Outstanding stage presence.	The Odd

4 rows in set (0.00 sec)

VIEW to display financier's invoice details:

```
CREATE OR REPLACE VIEW invoice_details_view AS
SELECT i.invoice_number,i.invoice_date, f.first_name AS financier_first_name, f.last_name
AS financier_last_name, snp.snpacks_name,
CONCAT('$', FORMAT(snp.snpacks_price, 2)) AS snpack_price,
CONCAT('$', FORMAT(SUM(snp.snpacks_price), 2)) AS invoice_total
FROM invoices i
JOIN financier f ON i.financier_id = f.financier_id
JOIN invoice_packages ip ON i.invoice_id = ip.invoice_id
JOIN sweet_note_packs snp ON ip.snpacks_id = snp.snpacks_id
GROUP BY i.invoice_number, i.invoice_date, f.first_name, f.last_name, snp.snpacks_name,
snp.snpacks_price
ORDER BY i.invoice_date ASC;
```

Query OK, 0 rows affected (0.02 sec)

Test:

```
SELECT * FROM invoice_details_view;
```

invoice_number	invoice_date	financier_first_name	financier_last_name	snpacks_name	snpack_price	invoice_total
812000-399	2025-08-05	Katherine	Sommers	Half Note Pack	\$250.00	\$250.00
812000-400	2025-08-05	Elijah	Garcia	Quarter Note Pack	\$168.00	\$168.00
812000-401	2025-08-10	Omar	Abdi	Whole Note Pack	\$450.00	\$900.00
812000-402	2025-08-13	Anastasia	Petrova	Whole Note Pack	\$450.00	\$450.00

4 rows in set (0.00 sec)

Implement Stored Program:

Automatically creates a new invoice with one or more Sweet Note Packs for a financier.

```
DELIMITER //
CREATE PROCEDURE create_invoice_with_packs (
  IN p_financier_id INT,
  IN p_invoice_number VARCHAR(100),
  IN p_due_days INT,
  IN p_snpacks_ids TEXT
)
BEGIN
  DECLARE v_invoice_id INT;
  DECLARE v_total DECIMAL(10,2) DEFAULT 0.00;
  DECLARE pack_id INT;

  START TRANSACTION;

  INSERT INTO invoices (
    financier_id, invoice_number, invoice_date, invoice_total,
    payment_total, credit_total, invoice_due_date, payment_date)
  VALUES (p_financier_id, p_invoice_number, CURDATE(), 0.00, 0.00, 0.00, DATE_ADD(CURDATE(),
    INTERVAL p_due_days DAY), NULL
  );

  SET v_invoice_id = LAST_INSERT_ID();

  WHILE LOCATE(',', p_snpacks_ids) > 0 DO
    SET pack_id = CAST(SUBSTRING_INDEX(p_snpacks_ids, ',', 1) AS UNSIGNED);
    INSERT INTO invoice_packages (invoice_id, snpacks_id) VALUES (v_invoice_id, pack_id);

    SET v_total = v_total + (
      SELECT snp.snpacks_price * snp.lessons_per_pack
      FROM sweet_note_packs snp
      WHERE snp.snpacks_id = pack_id
    );

    SET p_snpacks_ids = SUBSTRING(p_snpacks_ids, LOCATE(',', p_snpacks_ids) + 1);
  END WHILE;

  SET pack_id = CAST(p_snpacks_ids AS UNSIGNED);
  INSERT INTO invoice_packages (invoice_id, snpacks_id) VALUES (v_invoice_id, pack_id);
  SET v_total = v_total + (
    SELECT snp.snpacks_price * snp.lessons_per_pack
    FROM sweet_note_packs snp
    WHERE snp.snpacks_id = pack_id
  );

  UPDATE invoices
  SET invoice_total = v_total
  WHERE invoice_id = v_invoice_id;
  COMMIT;
END //
```

Query OK, 0 rows affected (0.03 sec)

```
CALL create_invoice_with_packs(
1,
'INV-1002',
30,
'1,2'
);
```

Query OK, 0 rows affected (0.03 sec)

Test stored program:

```
SELECT i.invoice_id, i.invoice_number, i.invoice_date,
i.invoice_due_date, f.first_name AS financier_first_name, f.last_name AS
financier_last_name, snp.snpacks_name, snp.snpacks_price,
snp.lessons_per_pack
FROM invoices i
JOIN financier f ON i.financier_id = f.financier_id
JOIN invoice_packages ip ON i.invoice_id = ip.invoice_id
JOIN sweet_note_packs snp ON ip.snpacks_id = snp.snpacks_id
WHERE i.invoice_number = 'INV-1002';
```

invoice_id	invoice_number	invoice_date	invoice_due_date	financier_first_name	financier_last_name	snpacks_name	snpacks_price	lessons_per_pack
5	INV-1002	2025-07-27	2025-08-26	Grace	Goodman	Quarter Note Pack	168.00	3
5	INV-1002	2025-07-27	2025-08-26	Grace	Goodman	Half Note Pack	250.00	5

2 rows in set (0.00 sec)